Human Factors and Ergonomics
EIN 3241

Class Periods: MWF, Period 9 (4:05 PM – 4:55 PM)
Location: FLG 0260

Instructor:
Name: Jad A. Atweh, Ph.D.
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Office Hours: Mondays and Wednesdays, Period 8 (3-3:50 PM; *or by appointment*)
Office Location: Weil 477

TAs/Graders:
Name: Makayla Shortino
Email Address: makaylashortino@ufl.edu
Office Hours: Mondays and Wednesdays, 9:30-10:30 AM
Location: Zoom (under the Zoom Conferences tab).

Name: Janna Scholtz
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Name: Teo Pedercini
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Office Hours: Tuesdays, 12:45-1:45 PM
Location: TBD

Course Description:
Introduces the techniques/concepts to understand users and workplace requirements for the design of sociotechnical systems. Topics covered include methods for work measurement, human cognitive and physical capabilities and limitations, and workplace requirements. Applications for design, including computer displays, noise, repetitive, and high physical effort tasks, are presented.

Course Pre-Requisites / Co-Requisites:
ESI 3215C (Data Analysis for Industrial Applications) with a minimum grade of C.

Course Objectives:
At the conclusion of this course, students will be able to:
- Analyze the role and importance of human factors and ergonomics in the design of safe and efficient systems.
- Evaluate human–machine system designs by considering human cognitive and physical capabilities and limitations.
- Select and correctly use appropriate work measurement, human-machine system analysis, and design tools
- Design and justify task modifications that enhance human performance and safety in workplace systems.
- Apply human-machine system design principles to real-world problems through exercises

Relation to Program Outcomes (ABET):
This course contributes to ensuring that the following program educational objectives of our BSISE program are met:
Within 5 years of graduation, BSISE graduates…
- Are successful professionals using industrial and systems engineering skills;
- Acquire advanced knowledge through continuing education or advanced degree programs;
- Are active leaders in their profession and/or community.

Outcome	Coverage
1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics	High
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	Medium
3. An ability to communicate effectively with a range of audiences	
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	Medium
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	Low
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	Medium

Recommended Materials

- Lee, J.D., Wickens, C. D., Liu, Y. D. & Boyle, L.N. (2017). Designing for People: An Introduction to Human Factors Engineering (3rd Ed.) ISBN: 978-1539808008.
- Fundamentals of Industrial Ergonomics, Second Edition, B. Mustafa Pulat, ISBN-13: 978-0881339499

Course Modules:

- Module 1: Introduction and History of Ergonomics
- Module 2: Design, Evaluation, and Research Methods
- Module 3: Engineering Anthropometry
- Module 4: Physical Ergonomics
- Module 5: Cognitive Ergonomics
- Module 6: Design of Displays and Controls
- Module 7: Work Measurement
- Module 8: Work Area and Job Design

Course Schedule (Tentative)

Week	*Date*	*Lecture Topic*	*Deliverables*	*Module*
1	Monday, January 12	Syllabus & Course Introduction		1
	Wednesday, January 14	Applications of Human Factors		
	Friday, January 16	History of Human Factors and Ergonomics		
2	Monday, January 19	**MLK Day (No Classes)**		
	Wednesday, January 21	Task Analysis & Design Methods		2
	Friday, January 23	Evaluation Methods		
3	Monday, January 26	Research Methods **(Quiz 1)**		
	Wednesday, January 28	Engineering Anthropometry		3
	Friday, January 30	Engineering Anthropometry		
4	Monday, February 2	Engineering Anthropometry		
	Wednesday, February 4	Lab 1: Engineering Anthropometry **(Quiz 2)**		
	Friday, February 6	Work Physiology		4
5	Monday, February 9	Work Physiology		

	Wednesday, February 11	Lab 2: EMG	*Lab 1*	
	Friday, February 13	Occupational Biomechanics		
6	Monday, February 16	Occupational Biomechanics		
	Wednesday, February 18	Lab 3: Strength **(Quiz 3)**	*Lab 2*	
	Friday, February 20	Visual Sensory System		5
7	Monday, February 23	Visual Sensory System		
	Wednesday, February 25	Auditory & Somatosensory Systems	*Lab 3*	
	Friday, February 27	Lab 4: Vision Screener **(Quiz 4)**		
8	Monday, March 2	Cognition and Human Information Processing		
	Wednesday, March 4	Cognition and Human Information Processing		
	Friday, March 6	Lab 5: Audiometer + Review Session	*Lab 4*	
9	Monday, March 9	**Midterm Exam (8:20 PM – 10:10 PM)**		
	Wednesday, March 11	Visual Search & Signal Detection Theory		
	Friday, March 13	Visual Search & Signal Detection Theory		
10	Monday, March 16	**Spring Break (No Classes)**		
	Wednesday, March 18			
	Friday, March 20			
11	Monday, March 23	Lab 6: Eye Tracking **(Quiz 5)**	*Lab 5*	5
	Wednesday, March 25	Design of Displays and Controls		6
	Friday, March 27	Design of Displays and Controls		
12	Monday, March 30	Time Studies		7
	Wednesday, April 1	Time Studies		
	Friday, April 3	Work Sampling	*Lab 6*	
13	Monday, April 6	Predetermined Motion Time Systems		
	Wednesday, April 8	Predetermined Motion Time Systems		
	Friday, April 10	Work Area Design		8
14	Monday, April 13	Job Design		
	Wednesday, April 15	Job Design **(Quiz 6)**		
	Friday, April 17	Project Presentations		
15	Monday, April 20	Project Presentations		
	Wednesday, April 22	Project Presentations		
	Thursday & Friday (April 23 & 24)	**Reading Days**	*Final Report (April 25)*	
16	Tuesday, April 28	**Final Exam (8:00 PM – 10:00 PM)**		

Grading Scheme – How will I demonstrate my knowledge and be assessed?

Assessment	Base Weights	Oops! I did poorly on the Midterm	I froze up on the Final	I struggled along the way but nailed the Final
In-Class Activities	5%	5%	5%	5%
Quizzes	15%	15%	15%	10%
Labs	10%	10%	10%	5%
Midterm Exam	25%	15%	35%	20%
Project	20%	20%	20%	20%
Final Exam	25%	35%	15%	40%

Grading Policy

Percent	Grade	Grade Points
93.4 - 100	A	4.00
90.0 - 93.3	A-	3.67
86.7 - 89.9	B+	3.33
83.4 - 86.6	B	3.00
80.0 - 83.3	B-	2.67
76.7 - 79.9	C+	2.33
73.4 - 76.6	C	2.00
70.0 - 73.3	C-	1.67
66.7 - 69.9	D+	1.33
63.4 - 66.6	D	1.00
60.0 - 63.3	D-	0.67
0 - 59.9	E	0.00

- A C is required to pass this course. A C– is **NOT** a passing grade.

Project Policies

- Peer evaluations will be conducted multiple times during the semester and will cover criteria such as attendance in team meetings, report writing, presentation delivery, and general work ethic.
- **Peer Evaluation grades may bump your grade up or down.** While the proposal and report receive a group grade, the individual grade may be modified based on individual contributions reported in peer evaluations.

Policies we hope never affect you! (not taking the exams, late work, team members not contributing…)

- Please note that the Midterm Exam date is scheduled at the beginning of the semester and announced in advance. This will give you time to make arrangements. Students are expected to prioritize this date.
- If health, disabilities, or family emergencies activities prevent you from turning in work on time, talk to me as soon as you can! Otherwise, no credit will be assigned to assignments submitted after the deadline.
- If a group member contributes minimally or not at all on a case, please email or talk to Dr. Atweh.
- We hold the Honor Code in high esteem! If you are caught cheating, your work will receive a zero and you will receive a failing grade in the course. This is separate from any pursuit of an Honor charge.

Commitment to a Safe and Inclusive Learning Environment

The Herbert Wertheim College of Engineering values varied perspectives within our community and is committed to supporting the University's core values, including the elimination of discrimination. It is expected that every person in this class will treat one another with dignity and respect regardless of race, creed, color, religion, age, disability, sex, sexual orientation, gender identity and expression, marital status, national origin, political opinions or affiliations, genetic information, and veteran status.

If you feel like your performance in class is being impacted by discrimination or harassment of any kind, please contact your instructor or any of the following:
• Your academic advisor or Undergraduate Coordinator
• HWCOE Human Resources, 352-392-0904, student-support-hr@eng.ufl.edu
• Pam Dickrell, Associate Dean of Student Affairs, 352-392-2177, pld@ufl.edu

University Academic Policies & Campus Resources

For official academic policies and student support resources, please refer to the University of Florida's syllabus policies: https://go.ufl.edu/syllabuspolicies. This includes information on attendance, grading, academic integrity, accommodations for students with disabilities, course evaluations, in-class recording policies, library services, and other valuable campus support.